

Week 10:

Requirements Analysis and Design Definition

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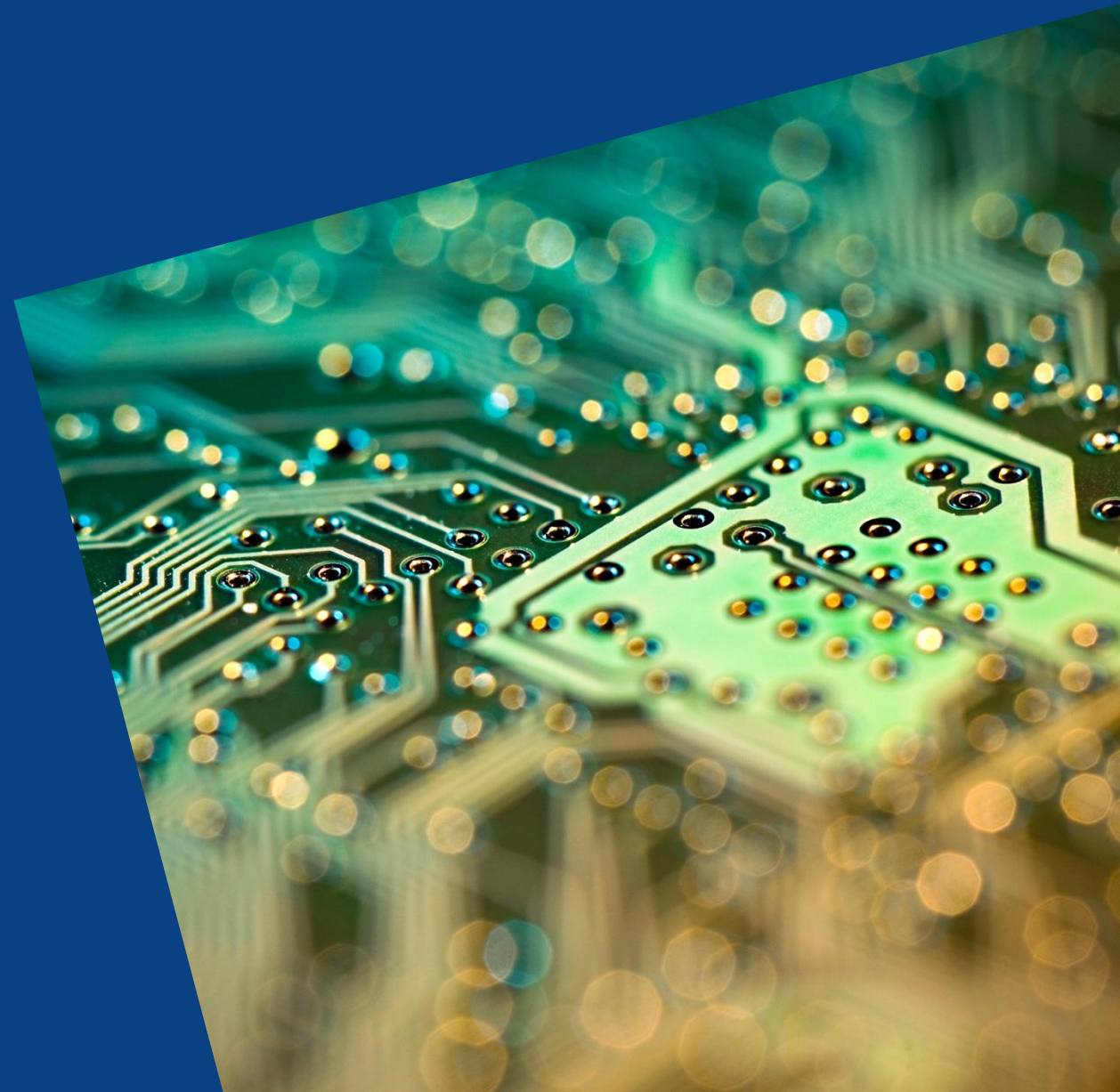
School of Computing and Information Systems





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Requirements Analysis and Design Definition (BABOK)





Knowledge area:

Requirements analysis & design definition

- Structure and organize requirements discovered during elicitation activities
- Specify and model requirements and designs
- Validate and verify information
- Identify solution options that meet business needs
- Estimate the potential value that could be realized for each solution option

Input/output diagram

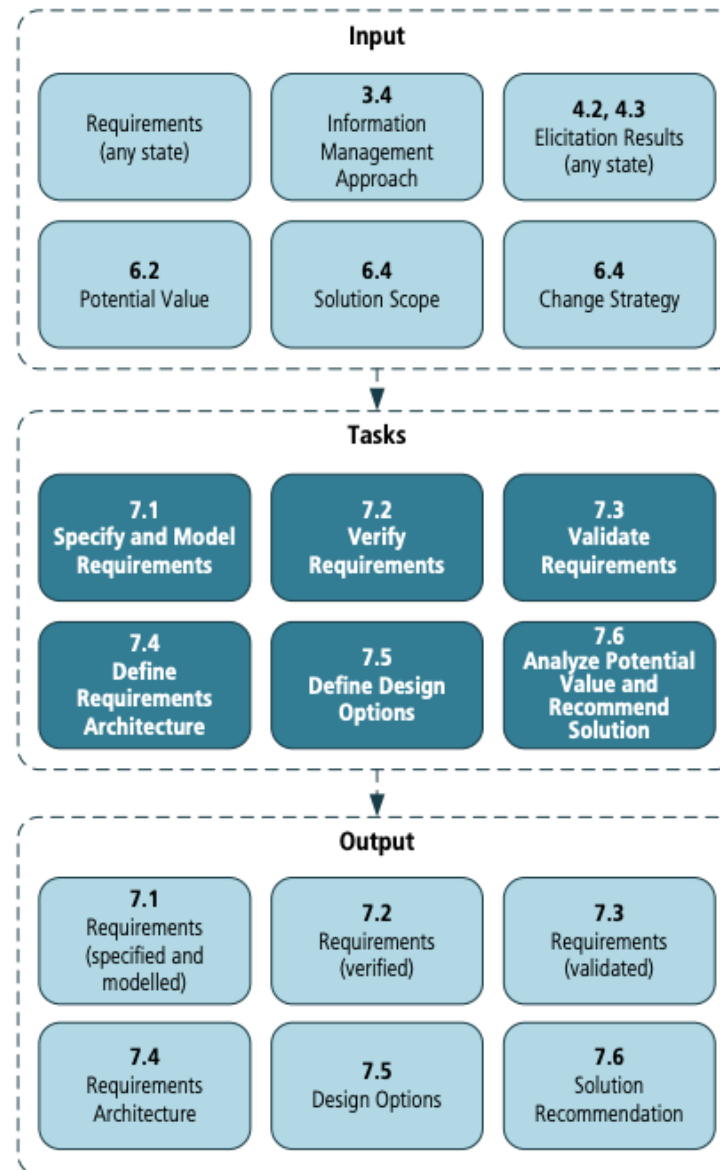
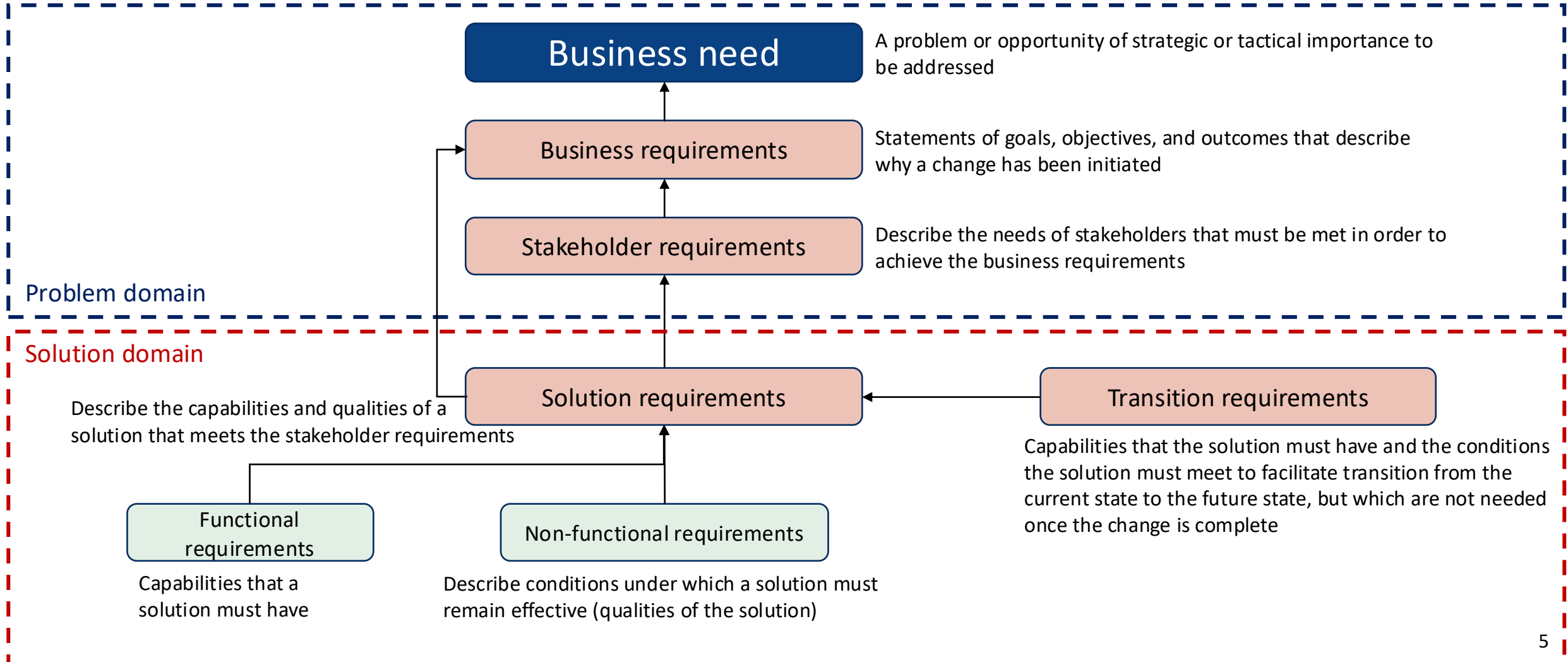


Figure 7.0.2 - BABOK

Requirements classification scheme



Functional vs. Non-Functional Requirements



Requirements and design cycle

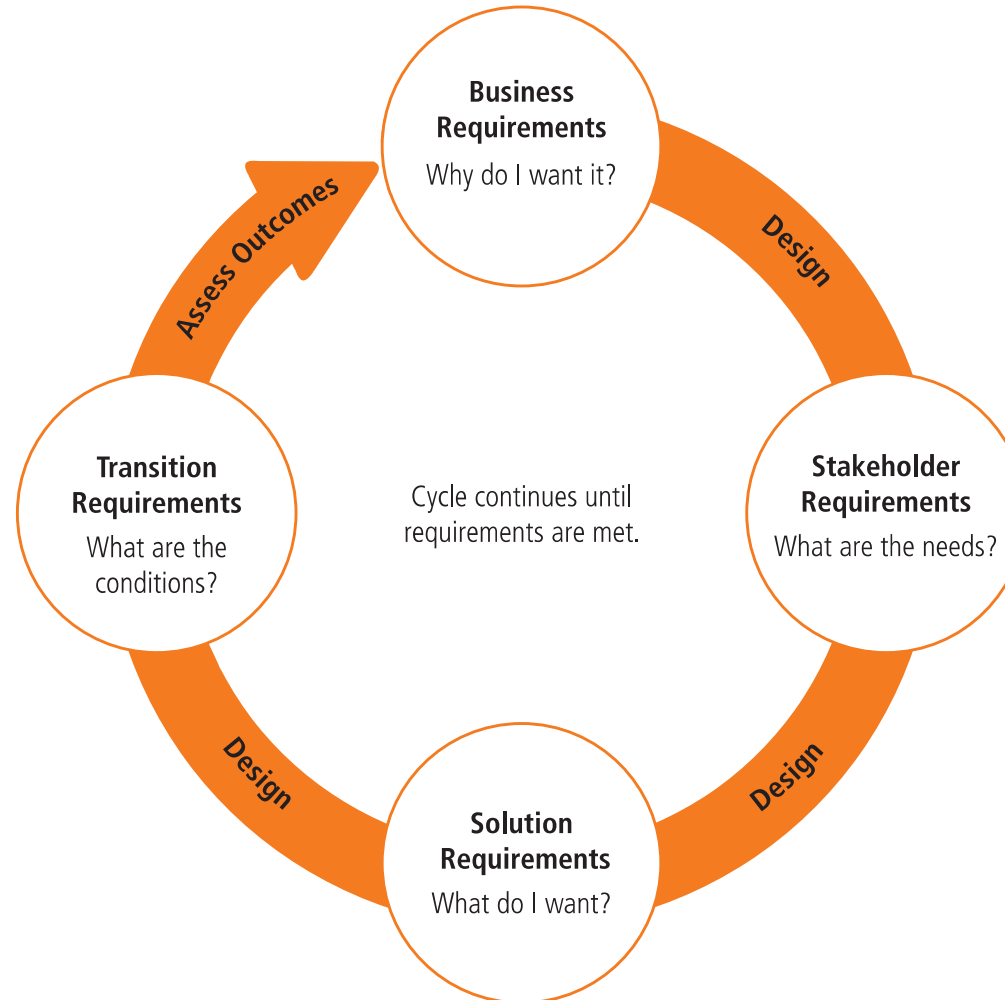


Figure 2.5.1 - BABOK

Requirement analysis & design definition: Tasks

Specify and model
requirements

Verify requirements

Validate
requirements

Define
requirements
architecture

Define design
options

Analyze potential
value and
recommend
solution

Requirement analysis & design definition: Tasks

Specify and model
requirements

Verify requirements

Validate
requirements

Analyze, synthesize and refine elicitation
results into requirements and designs

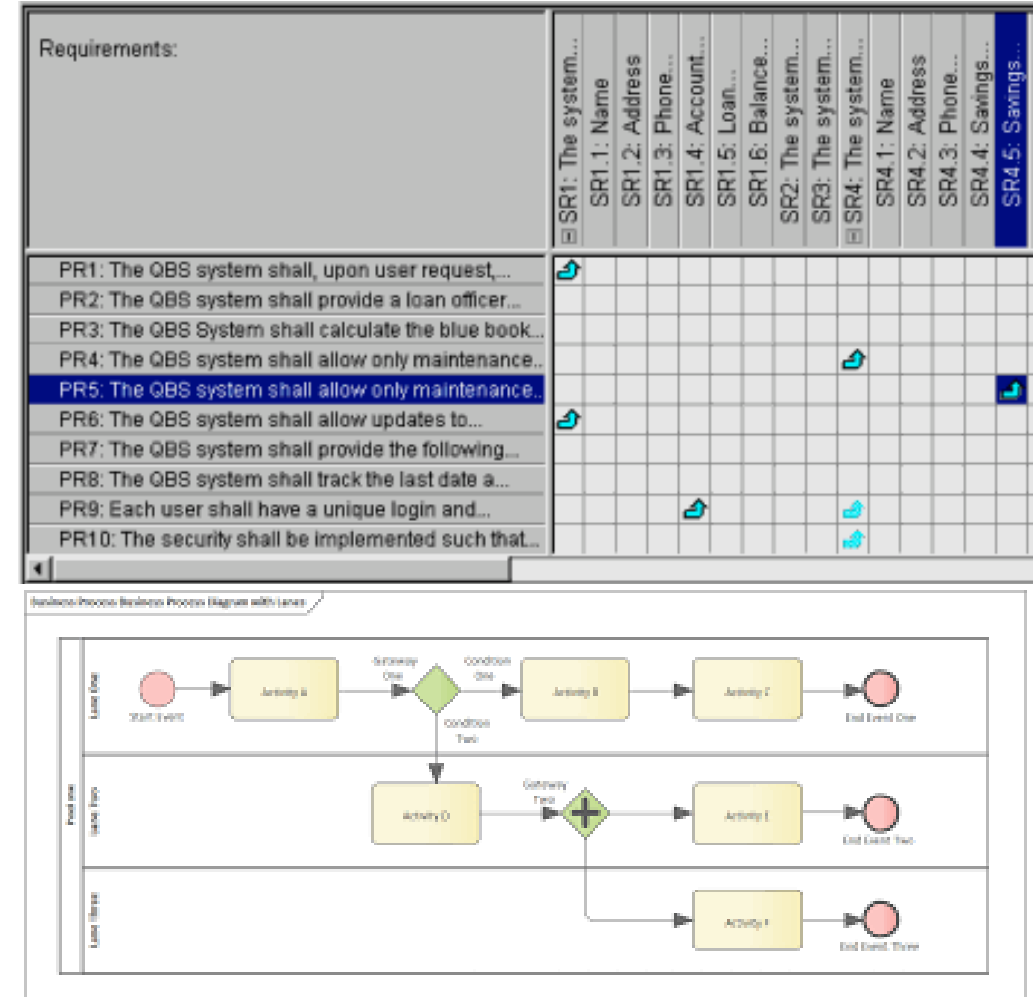


Specify and model requirements

- **Model requirements:** Textual or visual representations are created
- **Analyse requirements:** Assess if something is missing, unnecessary or has to be changed
- **Represent requirements and attributes:** document and represent all necessary information
- **Implement appropriate levels of abstraction:** assess levels of detail and representation formats. Keep in mind that different stakeholders require different levels of detail

Specify and model requirements: models

- Textual or visual representations are created
- The format of these representations has to be digestible by relevant stakeholder groups
- Common representations for requirements are:
 - Matrices
 - Diagrams (e.g., BPMN)



Specify and model requirements: model categories

- **People and roles:** represent organizations, groups of people, roles and their relationship. For example: personas or organizational models
- **Rationale:** represent the 'why' of a change. For example: decision modelling or root cause analysis
- **Activity flow:** sequences of actions, events or choices. For examples: process modelling (BPMN)
- **Capability:** features or functions of an enterprise or a solution. For example: prototyping
- **Data and information:** characteristics or exchange of information within an enterprise or a solution. For example: state modelling

Requirement analysis & design definition: Tasks

Specify and model
requirements

Verify requirements

Validate
requirements

Ensures that a set of requirements or designs has been developed in enough detail to be usable by a particular stakeholder, is internally consistent, and is of high quality



Verify requirements

- Ensure requirements and designs have been defined correctly (see next slide)
- Specifications and models must be correct (check notation), well written and easily understood
- Business analysts and stakeholders need to determine if all these elements are ready for validation



Verify requirements: characteristics of requirements

Atomic: It can be understood independently of other requirements or designs. It is self-contained and can stand on its own as a statement or expression.

Complete: The requirement must have enough detail for work to be able to continue. The level of completeness is not always the same.

Consistent: It is aligned with the identified needs of the stakeholders and does not contradict any other requirement.

Concise: Has enough information to describe itself. It doesn't contain unnecessary explanations or content.



Verify requirements: characteristics of requirements

Feasible: It is in line with the expectations of the project in terms of risk, schedule, budget or when it is considered worthy to be explored in more detail.

Unambiguous: It is stated in a clear and straightforward way that describes whether a solution does or does not meet the associated need.

Testable: It is clear to the reader whether a requirement or design has been fulfilled by a solution or not.

Prioritized: It shows its relative importance and/or value in relation to the other requirements it is associated with.

Understandable: It uses common terminology of the intended audience.

Requirement analysis & design definition: Tasks

Specify and model
requirements

Verify requirements

Validate
requirements

Ensures that all requirements and designs align
to the business requirements and support the
organization's goals and objectives



Validate requirements

- **Identify assumptions:** assumptions provide a basis for decision making where information is not available or for managing risks attached to a solution
- **Define measurable evaluation criteria:** criteria to evaluate how successful the change is after the solution is implemented
- **Evaluate alignment with solution scope:** ensure the solution is aligned with the stakeholder requirements

Activity

The **BrainFuel App** should be designed for you to...

-  ...track study hours and get coffee vouchers
-  ...scan your receipts for snack cashback
-  ...redeem points for energy drinks, fruit, or trail mix
-  ...participate in group challenges: Earn extra when studying in a group
-  ... see your stats on focus streaks + study time

When poll is active
respond at

[PollEv.com](https://pollev.com/julianmarx588)
[/julianmarx588](https://pollev.com/julianmarx588)



Requirement analysis & design definition: Tasks

Ensure that the requirements collectively support one another to fully achieve the objectives

Define
requirements
architecture

Define design
options

Analyze potential
value and
recommend
solution



Define requirements architecture

- **Requirements viewpoints and views:** define which model types, notations and attributes should be used for documenting the requirements
- **Template architectures:** standards used across industry, sector or organisation
- **Completeness:** ensure completeness
- **Relate and verify requirements relationships**
- **Business analysis information architecture:** define the relationship between requirements, designs, types of models and elicitation results

Requirement analysis & design definition: Tasks

Identifies, explores and describes different possible ways of meeting the need and reach the future state

Define
requirements
architecture

Define design
options

Analyze potential
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recommend
solution

Define design options

- **Defining solution approaches:** determining whether a solution component will be purchased, created or a combination of both
- **Identify improvement opportunities:** identify opportunities to improve business operations (e.g., increase efficiencies, identify access to information and identify additional capabilities)
- **Requirements allocation:** requirements are assigned to specific solution components or planned releases
- **Describe design options:** design options are described while considering the desired future state (they can include: business policies, processes, organisational structures, etc.)

Requirement analysis & design definition: Tasks

Estimate the potential value for each design option and establish which one is the most appropriate to meet the requirements

Define
requirements
architecture

Define design
options

Analyze potential
value and
recommend
solution



Analyze potential value and recommend solution

- **Expected benefits**
- **Expected costs** (e.g., time, effort, operating costs, maintenance costs, human resources, etc.)
- **Determine value:** determine whether the benefits outweigh the costs, or vice versa
- **Assess design options and recommend solution**

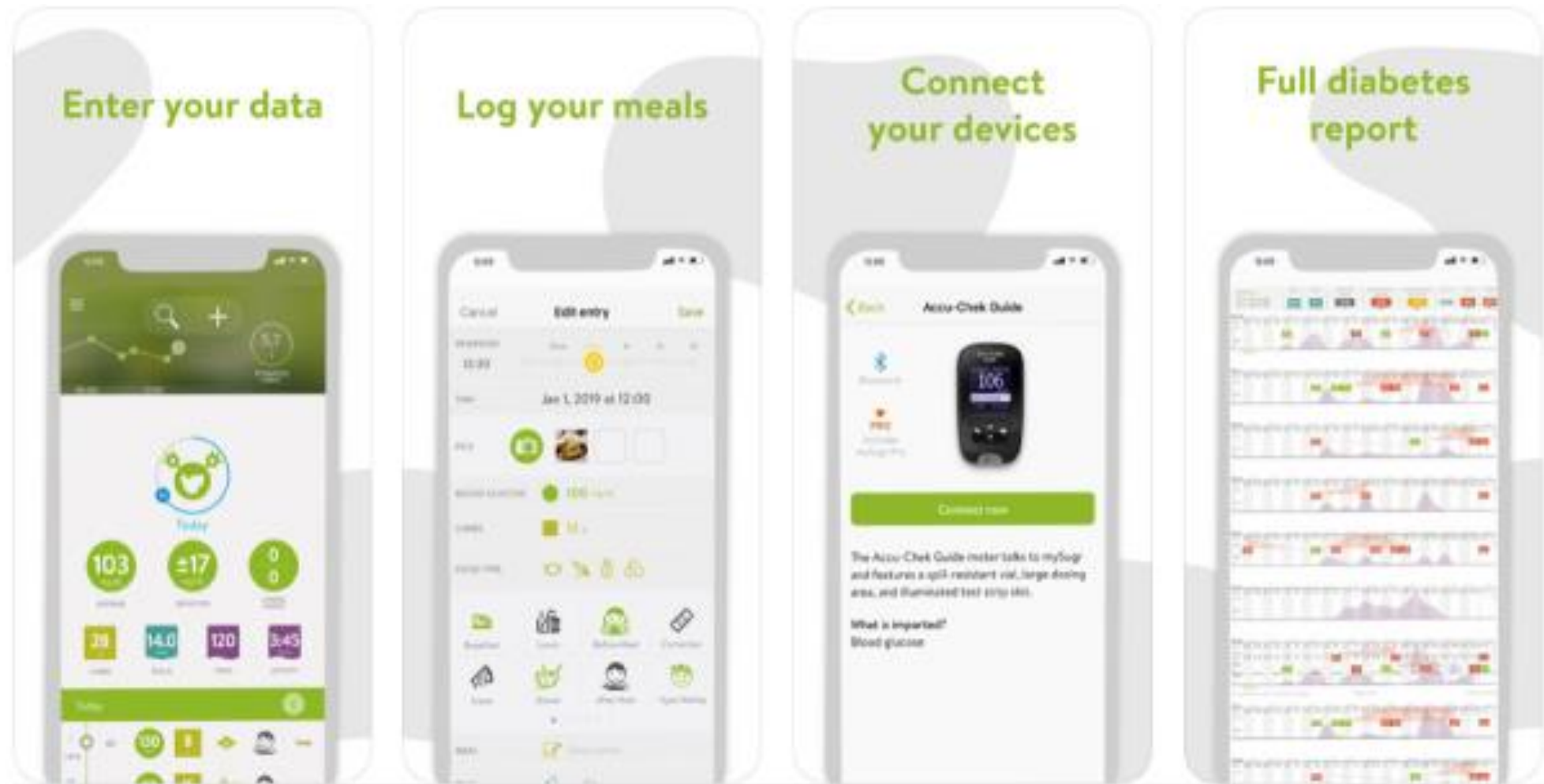


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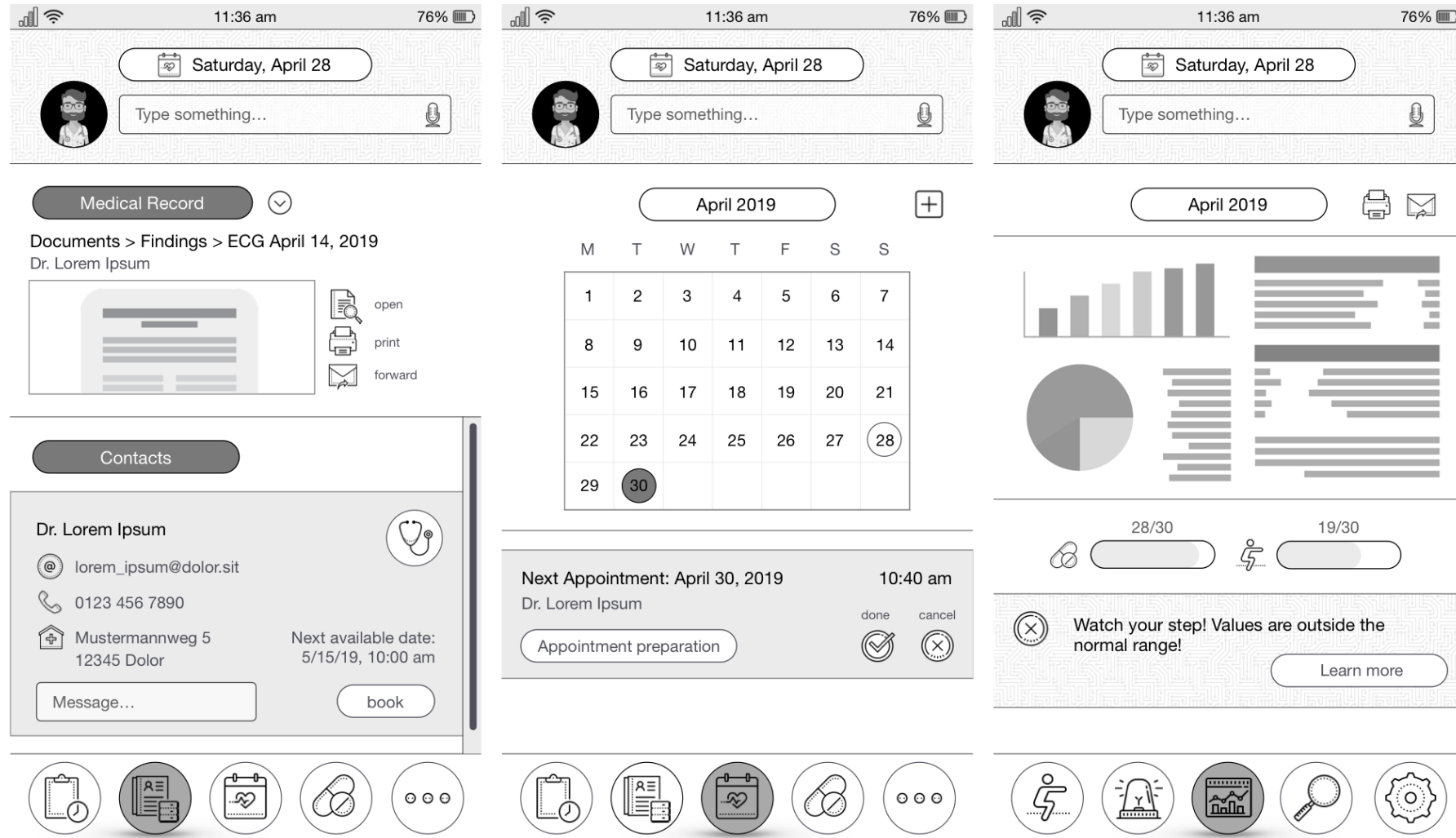
Wireframes



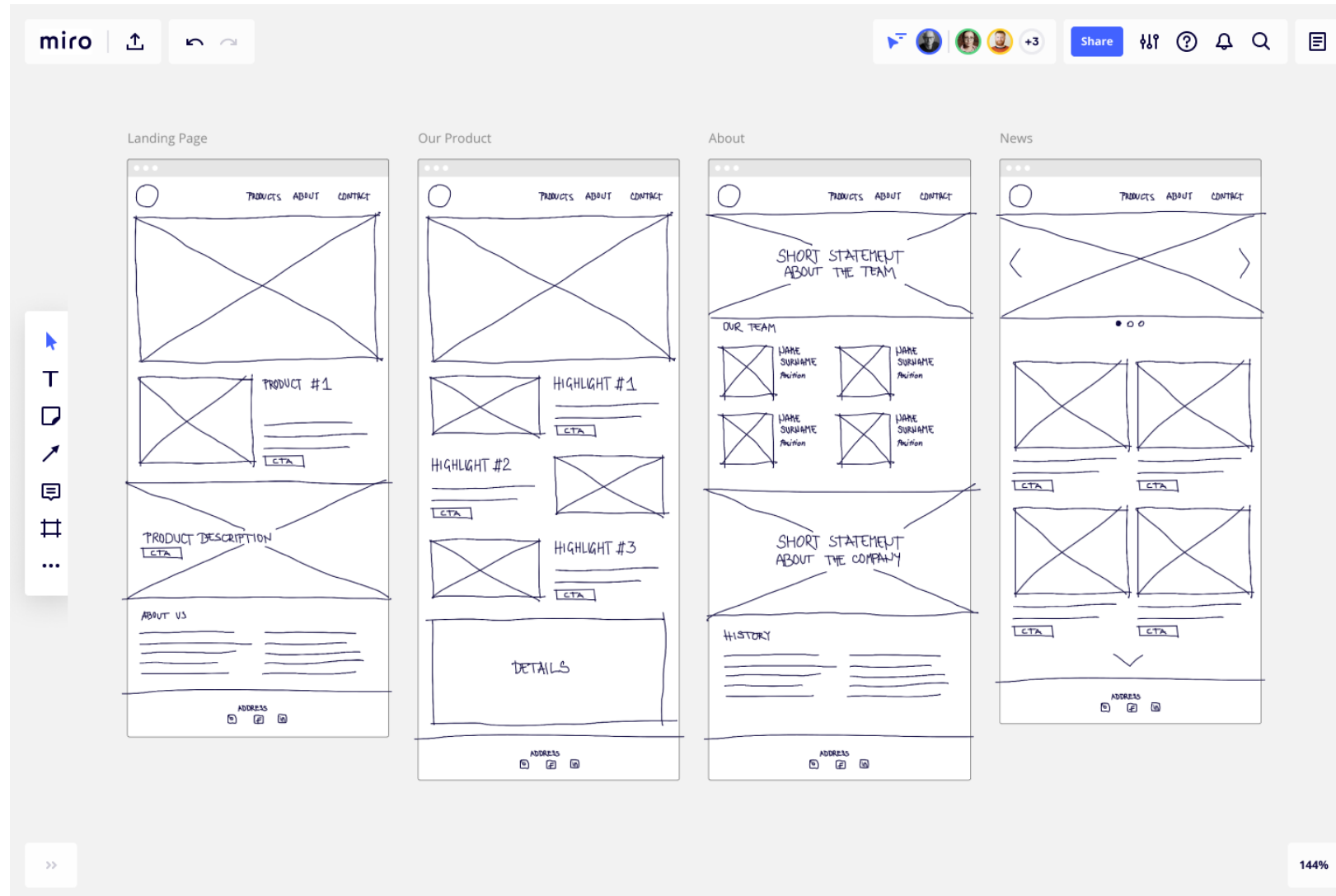
Example: high fidelity prototypes



Example: medium fidelity prototypes



Example: Low fidelity prototypes (wireframes)



<https://miro.com/templates/low-fidelity-wireframes/>



Wireframes

- A simplified representation of the system
- Can be hand-drawn or electronically generated
- Focus on structural elements of priority (no visuals or color)
- Low-fidelity prototypes (all wireframes are prototypes but not all prototypes are wireframes)
- Useful for idea generation, brainstorming and requirement analysis

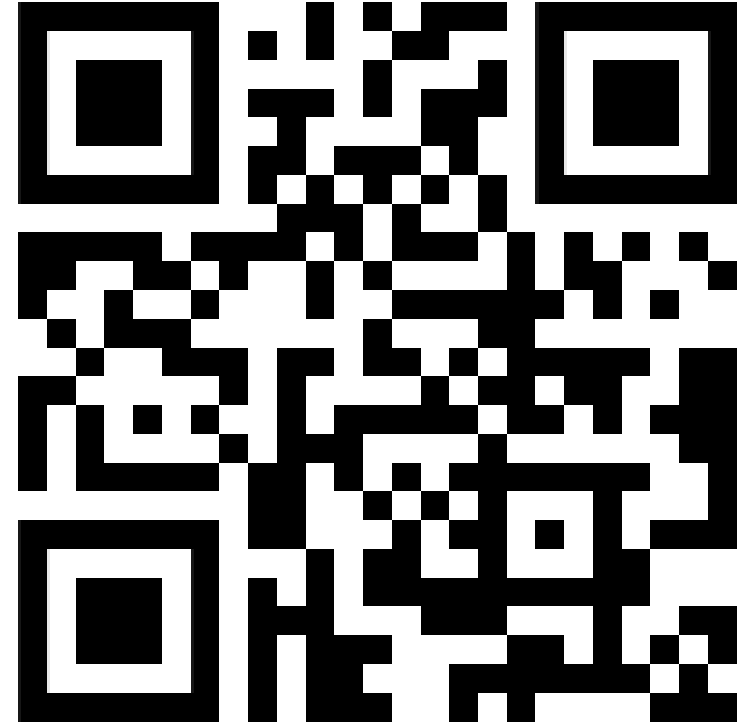
Wireframe

<https://wireframe.cc>

Activity:

Create a wireframe for a checkout page a customer sees when placing a pre-order (on a Desktop device).

Include at least a product image, fields for billing information, a pre-order button, and a money-back guarantee.





Wireframes

Yes, and...



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Use case diagrams





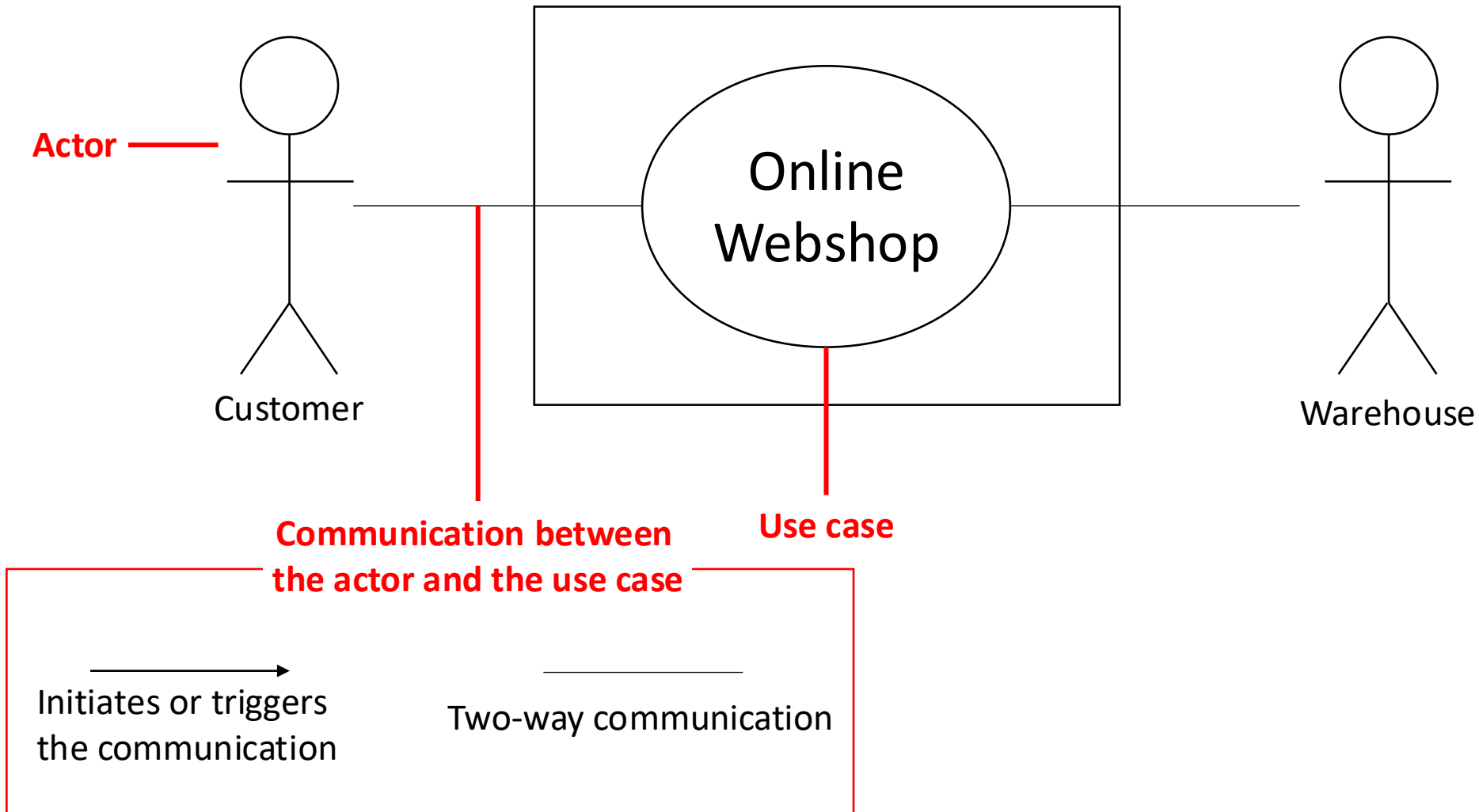
Use cases

- Illustrate how users interact with a system
- Are technology independent, so no IT knowledge needed
- Have a **use case narrative** that describes step-by-step the interaction of the user with the system
- Use cases can be modelled using use case diagrams in the Unified Modelling Language (UML)

Use cases: ingredients

- **Start & End:** When does a use case starts and when does it end?
- **Actors:** anyone or anything that interacts with the system
 - Stakeholders are anyone interested in the system but might not act in any of the use cases
- **Use cases:** functionality
- **Links** between actors and use cases: directed, undirected and can **extend**, **include** and **generalise**

Use cases: ingredients

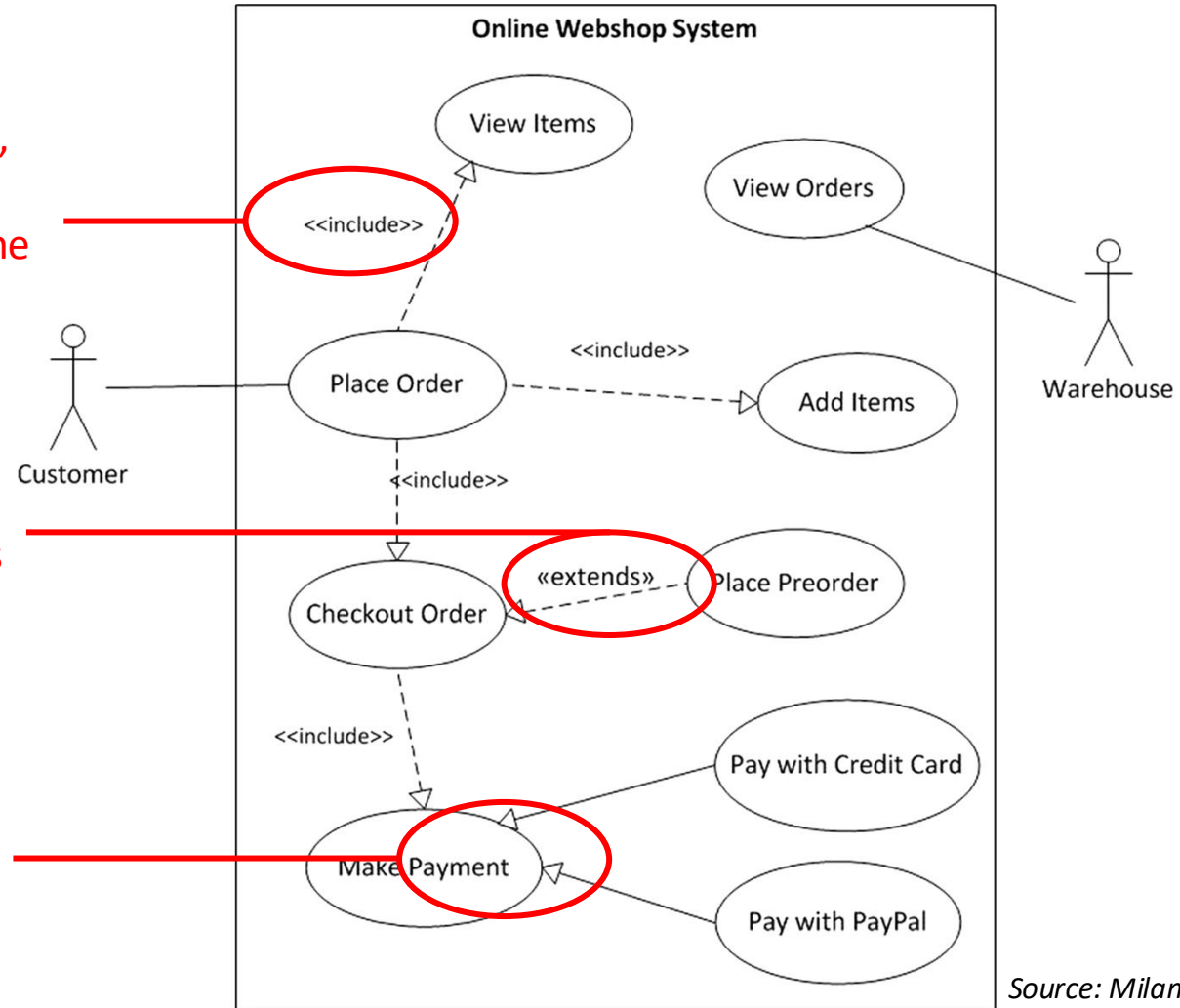


Use cases: example

Functionality of a use case,
uses or includes the
functionality of another one

If a specific condition is
fulfilled another use case is
needed

Subtypes

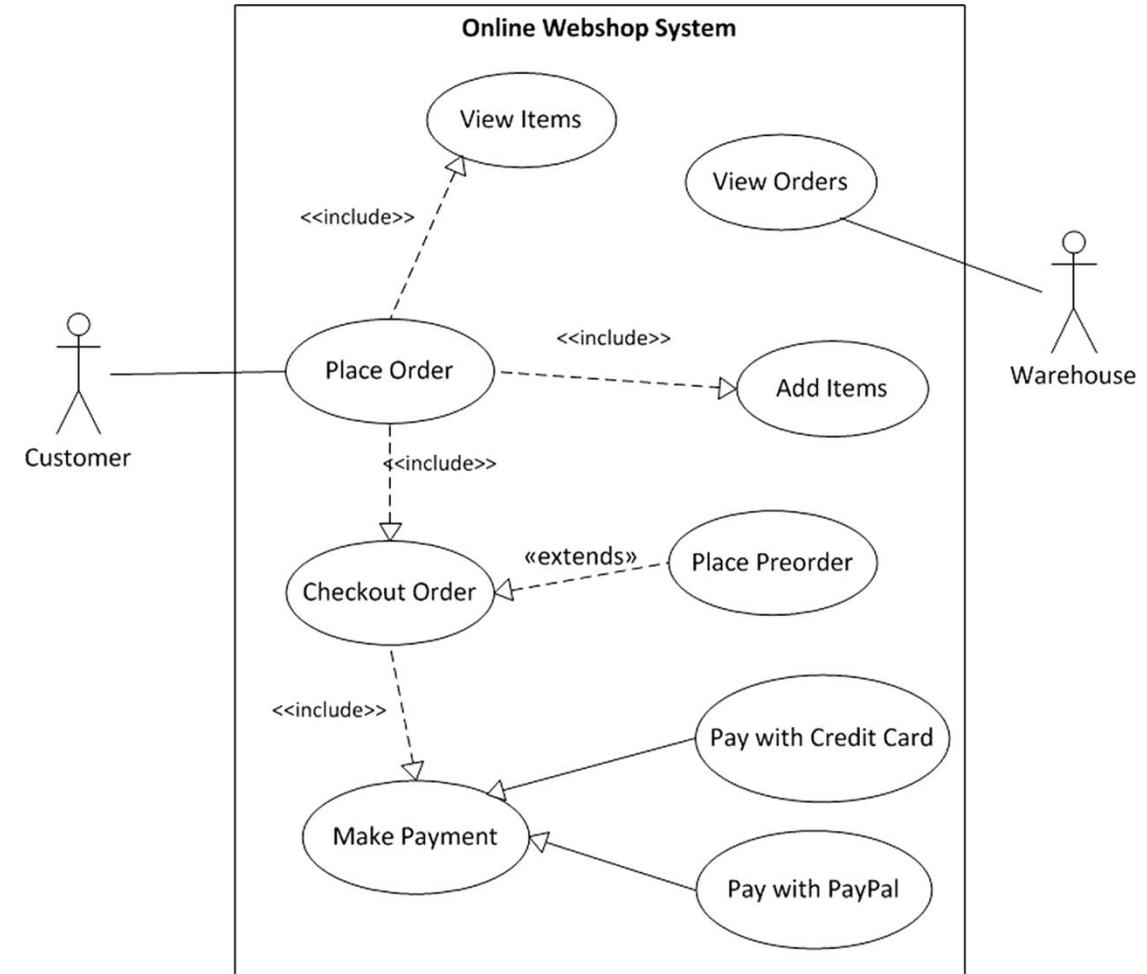


Source: Milani (2019)

Use case narrative example

- Actors: customer and warehouse

The customer will place an order after having viewed and added the items they want to buy. The customer will then proceed to checkout. If any item selected is not released yet, the customer will be asked if they wish to place a pre-order. If so, the customer will be informed of the estimated delivery time. The customer can then pay for the order with a credit card, in which case they fill out the necessary information. The customer can also choose to pay with PayPal. Once the payment is completed, the customer will receive an order and payment confirmation. The warehouse will view orders and fill them accordingly.



Full use case example

Actors	Registered customer (has an account and may give shipping and billing information (e.g., address))
	Warehouse (staff that fills the orders and sends it to the customer)
Triggers	The customer wants to buy the items
Pre-conditions	The customer has chosen the items they want to buy
Post-conditions	The order is placed in the system
	The customer receives an order confirmation with estimated delivery time
	The customer receives confirmation of payment made

Full use case: normal flow

1. The customer places the order
2. The system presents the shipping and billing information to the customer
3. The customer confirms the shipping and billing information for the order
4. The system shows: items, total and shipping costs
5. The customer confirms the order information
6. The system shows estimated delivery date
7. The customer makes the payment for the order
8. The system informs the customer of the order being finalised
9. The system forwards the order for fulfillment
10. The warehouse will fill the order and send it to the shipping address
11. The customer exits the system



Full use case: alternative flow

The customer wants to use a different shipping and billing address or update existing addresses

1. The customer chooses an option to use a different billing or shipping address
2. The customer enters the desired shipping and shipping information
3. The system updates the order with the new shipping and billing address
4. The use case continues

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